

Understanding the Mathematics of Letter III and the Theory of Entropicity

What the whole program is trying to do

The fastest way to understand Letter III is to separate it into three layers. The first layer is **standard mathematics**: manifolds, tangent spaces, tensors, Fisher–Rao geometry, Fubini–Study geometry, Kullback–Leibler divergence, quantum relative entropy, and the Bianchi identity. The second layer is **standard physics**: the Einstein–Hilbert action, curvature, geodesics, and the stress–energy tensor. The third layer is **ToE-specific structure**: the Obidi transformation, the Obidi metric, the Obidi Curvature Invariant, the Hybrid Metric–Affine Space, the irreversible/reversible split of the entropic action, and the proposal that spacetime and matter are both coarse-grained forms of information. Letter III explicitly lays out this program as a staged emergence from information geometry to effective gravity. ¹

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A very short slogan is this:

probability geometry $\longrightarrow$ quantum geometry $\longrightarrow$ Lorentzian spacetime geometry $\longrightarrow$ Einstein gravity.

In standard information geometry, the “distance” between nearby probability distributions is measured by the Fisher–Rao metric, and the mismatch between two full probability models is measured by divergences such as the Kullback–Leibler divergence. In quantum theory, the analogous geometry for pure states is the Fubini–Study metric, and for mixed states one uses quantum Fisher/Bures-type structures and quantum relative entropy. Letter III proposes that these are not just useful statistical tools but the deep pre-geometric objects from which physical spacetime is constructed. That is a bold and nonstandard claim, and it is important to treat it as a **theory proposal** rather than a settled textbook theorem. ²

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For a first-year high school student, the simplest intuitive picture is this. Imagine that reality is not built from “solid little things” first, but from patterns of distinguishability: what can be told apart from what, how strongly, and with what cost in entropy. If those patterns are very smooth, they look like a curved geometric space. If one direction in that geometry is singled out by entropy flow, that direction can behave like **time**. If concentrated information clusters move and interact inside that emergent geometry, they can appear to us as **matter, radiation, pressure, and stress**. That is the central intuition behind ToE. ³

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The geometry words you need first

A **manifold** is a space that may be curved globally but looks flat up close. The surface of Earth is the classic example: from far away it is round, but from where you stand it looks almost like a flat sheet. In math, a manifold is the natural stage on which calculus can be done. A **tangent space** $T_p M$ at a point p is the set of all possible tiny directions you could move from p . A **cotangent space** $T_p^* M$ is the dual space: it consists of

all linear measuring rules that take one tangent direction and return one number. If a vector is an arrow, a covector is a ruler. The total collection of all tangent spaces is the **tangent bundle** TM , and the total collection of all cotangent spaces is the **cotangent bundle** T^*M . These are examples of **vector bundles**, and bundles more generally are spaces that locally look like “base space $\times$ fiber.” 4

You can write those bundles schematically as

$$TM = \bigsqcup_{p \in M} T_p M, \quad T^*M = \bigsqcup_{p \in M} T_p^* M.$$

The symbol $\bigsqcup$ means “put all of them together, but keep track of which point they belong to.” A **fiber** is the little space sitting above each point. For the tangent bundle, the fiber above p is $T_p M$. For ToE, this “fiber picture” matters because Letter III wants ordinary spacetime to be the visible **base**, while hidden information or entropic microstructure can live in additional fibers that later get coarse-grained away. 5

A **scalar** is just one number at each point, like temperature $T(x)$ or entropy density $S(x)$. A **vector** is a directional quantity, like a velocity field $v^\mu(x)$. A **covector** or **one-form** is something like dS , which tells you how fast entropy changes in each direction. A **tensor** is a multilinear machine. A rank-2 tensor takes two vectors or covectors and returns a number. Important rank-2 tensors include the metric $g_{\mu\nu}$, the covariance matrix in statistics, and the stress-energy tensor $T_{\mu\nu}$ in relativity. If you remember only one thing, remember this: **the metric measures, the connection compares, and the curvature tells you whether going around a loop changes things.** 6

The **gradient** needs one careful remark because students often meet it in two slightly different disguises. The most primitive object is the differential

$$dS = \partial_\mu S dx^\mu,$$

which is a covector. Once a metric exists, you can raise the index and turn it into a vector:

$$(\nabla S)^\mu = g^{\mu\nu} \partial_\nu S.$$

So in curved geometry, “the gradient” is first a covector and only becomes a vector after the metric is used. This matters in Letter III because the Obidi transformation uses an entropy-gradient direction to build a timelike axis. 7

A **pullback** and a **pushforward** tell you how geometry moves along a map. If $\phi : M \rightarrow N$ is a smooth map, then the pushforward $d\phi_x : T_x M \rightarrow T_{\phi(x)} N$ sends arrows on M to arrows on N . The pullback does the opposite kind of job for covectors and tensors. If H_{AB} is a metric on an “information manifold” $\mathcal{I}$, then an emergent spacetime metric on M can be written as a pullback:

$$g_{\mu\nu}(x) = \partial_\mu \phi^A(x) \partial_\nu \phi^B(x) H_{AB}(\phi(x)).$$

This one formula captures a huge amount of the spirit of Letter III: spacetime geometry is not primitive, but inherited from a deeper information geometry. 7

A **fiber integral** is the mathematical version of “average over hidden details.” If $\pi : E \rightarrow B$ is a bundle and each fiber $F_b = \pi^{-1}(b)$ carries hidden variables, then an effective field on the base can be obtained by integrating over the fiber:

$$f_{\text{eff}}(b) = \int_{F_b} f(b, y) d\mu_{F_b}(y).$$

Physically, this is how coarse-graining works. You do not track every microscopic degree of freedom; you integrate them out. In ToE language, this is exactly the kind of move needed to go from a deep information manifold to effective four-dimensional physics. ⁷

The language of **moments** is also essential. If $p(x)$ is a probability density, then

$$\int p(x) dx = 1$$

is the **zeroth moment** and says total probability is normalized. The **first moment**

$$\mu = \int x p(x) dx$$

is the mean or center. The **second moment**

$$M_2 = \int x^2 p(x) dx, \quad \sigma^2 = \int (x - \mu)^2 p(x) dx$$

measures spread or variance. The **third moment** measures asymmetry or skewness. Why does this matter? Because many physical tensors are really moment objects. In kinetic theory and radiation theory, stress-energy and radiation tensors are moments of a distribution function. So your intuition that mass, pressure, radiation, and stress are “organized information” is mathematically sound in spirit: they are coarse-grained moment-patterns of underlying distributions. ⁸

The information-geometry engine

Now we come to the heart of information geometry. A **statistical manifold** is a family of probability distributions $p(x | \theta)$, where the parameters $\theta = (\theta^1, \dots, \theta^n)$ act like coordinates on a geometric space. The central metric on this space is the **Fisher-Rao metric**:

$$g_{ij}(\theta) = \mathbb{E}_\theta[\partial_i \log p(X | \theta) \partial_j \log p(X | \theta)].$$

You can also write it, under regularity assumptions, as minus the expected Hessian of the log-likelihood. The physical meaning is simple: it measures how easy it is to distinguish two nearby probability models. If a tiny parameter change causes a big change in the distribution, the Fisher metric is large. If the distribution barely changes, the metric is small. This links geometry directly to distinguishability and to the Cramér-Rao bound in statistics. ⁹

A very concrete example is a biased coin with probability q of heads. The distribution is

$$p(x | q) = q^x(1 - q)^{1-x}, \quad x \in \{0, 1\}.$$

Its log-likelihood is

$$\ell = x \ln q + (1 - x) \ln(1 - q),$$

so

$$\partial_q \ell = \frac{x}{q} - \frac{1-x}{1-q}.$$

Taking the expectation gives

$$g(q) = \mathbb{E}[(\partial_q \ell)^2] = \frac{1}{q(1-q)}.$$

This is a wonderful formula for intuition. Near $q = 1/2$, changing q a little changes the model moderately. But near $q = 0$ or $q = 1$, the same tiny change matters much more because the coin is almost locked into one behavior. So the geometric “sensitivity” blows up near the edges. That is what the Fisher metric is telling you. ¹⁰

The main divergence behind classical information geometry is the **Kullback–Leibler divergence**:

$$D_{\text{KL}}(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

or, in the discrete case,

$$D_{\text{KL}}(P||Q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

This is not an ordinary distance because it is not symmetric:

$$D_{\text{KL}}(P||Q) \neq D_{\text{KL}}(Q||P).$$

Its meaning is “how much extra coding cost or descriptive error you pay if the truth is P but you insist on using Q .” The reason KL divergence is central is that its local second-order expansion produces the Fisher metric:

$$D_{\text{KL}}(p_{\theta+d\theta}||p_{\theta}) = \frac{1}{2} g_{ij}(\theta) d\theta^i d\theta^j + O(d\theta^3).$$

So the Fisher metric is the infinitesimal quadratic shadow of KL divergence. ¹¹

One of the deepest results in the field is **Čencov’s theorem**. Roughly speaking, if you ask for a Riemannian metric on statistical models that behaves naturally under sufficient statistics and information-preserving transformations, then the Fisher metric is essentially the only answer, up to an overall constant. That is why Fisher–Rao is not just one convenient metric among many; it is mathematically privileged. For ToE, this means Fisher–Rao is an excellent candidate for the deepest classical information metric. But it also creates a problem: Fisher–Rao is positive-definite, while relativistic spacetime metrics are Lorentzian and must have

one direction of opposite sign. Letter III answers that problem with additional structure, not by denying Čencov’s theorem. ¹² [filecite turn0file0](#)

The next object is the **Amari–Čencov cubic tensor**, often written as

$$C_{ijk} = \mathbb{E}_\theta[\partial_i \ell \partial_j \ell \partial_k \ell], \quad \ell = \log p(x | \theta).$$

This is like a third moment of the score function. If the Fisher metric is the second-order part of local statistical shape, then C_{ijk} measures the first non-quadratic asymmetry of that shape. It is one of the reasons information geometry is richer than ordinary Riemannian geometry. ¹³

Using the cubic tensor, one defines the **Amari–Čencov α -connections**. Different books use slightly different sign conventions, but the idea is the same: there is not just one natural way to define “straight lines” in probability space. Instead there is a one-parameter family,

$$\nabla^{(\alpha)},$$

with $\alpha = 0$ giving the Levi-Civita connection of the Fisher metric, $\alpha = 1$ related to exponential-family geometry, and $\alpha = -1$ related to mixture geometry. In one common convention,

$$\Gamma_{ijk}^{(\alpha)} = \Gamma_{ijk}^{(0)} - \frac{\alpha}{2} C_{ijk}.$$

The pair $\nabla^{(\alpha)}$ and $\nabla^{(-\alpha)}$ are dual to each other with respect to the Fisher metric. The simplest way to picture this is that probability space has two natural notions of straightness: one adapted to “multiply and normalize,” the other adapted to “mix and average.” Letter III uses this family of connections as part of its entropic dynamics and interprets nonzero α as encoding a more general, possibly irreversible informational transport law. That interpretation is ToE-specific, but the underlying geometry is standard. ¹⁴

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The quantum-information engine

Classical probabilities are not enough for Letter III, because quantum theory carries phase, superposition, and entanglement. The geometric space of pure quantum states is not the Hilbert space itself, but the space of **rays** in Hilbert space, because overall phase does not matter physically. The natural metric on this ray space is the **Fubini–Study metric**. For a normalized state $|\psi\rangle$, it is

$$ds^2 = \langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2.$$

This removes the physically meaningless phase direction and keeps only true distinguishability between nearby pure states. In quantum geometry this comes from the **quantum geometric tensor** $Q_{\mu\nu}$, whose real part is the metric and whose imaginary part is the Berry curvature. ¹⁵

For a single qubit, you can make this very visual. A pure qubit state can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

Then the Fubini–Study line element becomes

$$ds^2 = \frac{1}{4} (d\theta^2 + \sin^2 \theta d\phi^2) .$$

That is just the metric of a sphere, up to scale. So pure qubit states live on the Bloch sphere. This is a beautiful example of how “state space” becomes literal geometry. ¹⁶

For mixed states, the natural geometry is more subtle. The **Bures metric** and related quantum Fisher structures generalize Fubini–Study to density matrices ρ . A key fact is that the mixed-state geometry reduces to the Fubini–Study metric for pure states or in the appropriate zero-temperature limit. So there is a quantum-information ladder: classical Fisher geometry for probability distributions, Fubini–Study for pure quantum states, and Bures/quantum Fisher geometry for mixed quantum states. Letter III’s Hybrid Metric–Affine Space is explicitly designed to combine classical Fisher–Rao and quantum Fubini–Study ingredients into one pre-spacetime structure. ¹⁷ [filecite turns out file 0](#)

The quantum analogue of KL divergence is the **Umegaki quantum relative entropy**:

$$S(\rho||\sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)] .$$

If ρ and σ commute, then they can be diagonalized together and this reduces exactly to the classical KL divergence of their eigenvalue distributions. So quantum relative entropy is not a random new object; it is the genuine operator-level extension of classical relative entropy. It is nonnegative, vanishes only when the two states agree, and is monotone under physically allowed coarse-graining operations. ¹⁸

You asked specifically about **Araki–Umegaki divergence**. The cleanest way to say it is this. In finite-dimensional quantum information, the formula above is usually called **Umegaki relative entropy**. In operator-algebraic and quantum-field-theoretic settings, Araki developed the more general relative entropy framework for states on von Neumann algebras, and there are also Araki–Uhlmann formulations in QFT. So when people loosely say “Araki–Umegaki,” they usually mean the family resemblance between the finite-dimensional Umegaki formula and its operator-algebraic generalizations. For learning purposes, the formula above is the one you should hold onto first. ¹⁹

This matters for ToE because Bianconi’s recent “gravity from entropy” theory uses a **quantum relative entropy** between a spacetime metric operator and a matter-induced metric operator as the action. Letter III places ToE in conversation with that program but wants to go further, by making the geometry itself arise from information manifolds before the entropic gravitational action is written down. ²⁰

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The Einstein-gravity side

Once a metric $g_{\mu\nu}$ exists on spacetime, differential geometry gives you a chain of constructions. The metric determines the Christoffel connection

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\mu\sigma} - \partial_{\sigma} g_{\mu\nu}) ,$$

which tells you how vectors change when transported. From the connection you build the Riemann curvature tensor,

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.$$

Contracting indices gives the Ricci tensor $R_{\mu\nu}$, and contracting once more gives the scalar curvature R . Finally, the particular rank-2 tensor that enters Einstein's equation is

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

This is the left-hand side of general relativity. ²¹

The reason $G_{\mu\nu}$ is so special is the **Bianchi identity**. At the full curvature level one has a cyclic identity for the Riemann tensor, and when indices are contracted the result is

$$\nabla^\mu G_{\mu\nu} = 0.$$

This is not a matter equation; it is a pure geometric identity. Physically, it means the left-hand side of Einstein's equation is automatically divergence-free. Therefore the right-hand side must also be divergence-free. That is why matter must obey a local conservation law

$$\nabla^\mu T_{\mu\nu} = 0.$$

Any theory that wants to replace Einstein's matter source by an entropic source must satisfy the same structural requirement. This is one of the strongest mathematical constraints on ToE's right-hand side program. ²²

The metric field equations come from the **Einstein-Hilbert action**

$$S_{\text{EH}} = \frac{1}{2\kappa} \int R \sqrt{-g} d^4x, \quad \kappa = \frac{8\pi G}{c^4}.$$

Varying this action with respect to the metric gives Einstein's equations. The Palatini identity controls the variation of the Ricci tensor, and integration by parts moves derivative terms off the metric variation. That is why the Einstein tensor appears naturally rather than by guesswork. Letter III repeatedly uses this variational picture as the target low-energy limit of ToE. ²³ [filecite00turn0file001](#)

On the right-hand side stands the **stress-energy tensor** $T_{\mu\nu}$, which packages energy density, momentum density, pressure, shear stress, and flux. In a perfect fluid one writes

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu},$$

up to signature convention. Here ρ is energy density, p is pressure, and u^μ is the fluid four-velocity. The component T_{00} is energy density, T_{0i} is momentum density or energy flux, and T_{ij} encodes stresses and pressure. In statistical mechanics, stress-energy is deeply tied to distributions, entropy currents, and moments, so the user's intuition that pressure, radiation, and stress are "organized information" is mathematically compatible with mainstream covariant statistical mechanics even before one adopts ToE's stronger interpretation. ⁸

This gives the clean conceptual bridge to ToE. In ordinary general relativity, the right-hand side is not just “mass.” It is a rank-2 bookkeeping device for all localized flows and densities of physical content. In ToE, the proposal is that this physical content should be re-read as stabilized information on an entropic manifold. So mass is persistent stored information, radiation is propagating information, pressure is the response of local information density to compression, and stress is directional information transfer. That is a philosophical claim, but it is not mathematically crazy: tensors already describe organized moments and flows, and statistical mechanics already teaches us that such quantities emerge from coarse-grained distributions. ²⁴ [\[filecite turn0file0\]](#)

The ToE bridge from information manifold to spacetime

Here we reach the part that is original to Letter III. The **Hybrid Metric–Affine Space** of ToE is not a standard textbook object. Letter III introduces it as a combined information-geometric arena that mixes a classical Fisher–Rao sector with a quantum Fubini–Study sector and allows affine data beyond an ordinary Riemannian metric. In plain language: ToE does not want just one kind of informational geometry, because the world contains both classical statistics and quantum distinguishability. HMAS is the proposed stage on which both are present before effective spacetime emerges. ²⁵ [\[filecite turn0file0\]](#)

Letter III also introduces an α - q **constitutive relation**. Standard nonextensive statistical mechanics uses a deformation parameter q , as in Tsallis and Rényi-type entropies, while information geometry uses the connection parameter α . There is real literature relating q -deformations and generalized dual geometries, but the exact constitutive law used in ToE is a theory choice of Letter III, not a universally accepted theorem. The important intuition is that $q \nearrow 1$ means the information theory is no longer purely Shannon-like, and $\alpha \nearrow 0$ means the geometry is no longer purely self-dual Levi-Civita. ToE ties these together so that nonextensive entropy and nontrivial information transport become aspects of one structure. ²⁶ [\[filecite turn0file0\]](#)

The next move is the **emergence map** from the information manifold to spacetime. In its cleanest mathematical form, ToE wants something like

$$g_{\mu\nu}(x) = \partial_\mu \phi^A(x) \partial_\nu \phi^B(x) H_{AB}(\phi(x)),$$

where H_{AB} is some deeper information metric and ϕ tells us how observed spacetime points sit inside the informational arena. If hidden micro-information lives in extra fibers, then an effective spacetime action is obtained by fiber integration or coarse-graining. This is the precise meaning of the phrase “information geometry yields physical spacetime.” It does not mean information somehow magically becomes geometry; it means a pushforward/pullback/coarse-graining pipeline is defined. [\[filecite turn0file0\]](#)

The most distinctive ToE object is the **Obidi transformation**. According to Letter III, one starts with a positive-definite information metric h_{AB} and an entropy-gradient direction

$$u_A = \frac{\nabla_A S}{\|\nabla S\|_h}.$$

Then one defines the **Obidi metric** by a rank-one sign-flip:

$$g_{AB}^{(O)} = h_{AB} - 2u_A u_B,$$

possibly with an overall conformal factor in some versions of the construction. This is one of the cleanest ideas in the whole manuscript. Why? Because you can prove in one line that it creates a Lorentzian sign. Since u is unit with respect to h ,

$$g^{(O)}(u, u) = h(u, u) - 2[h(u, u)]^2 = 1 - 2 = -1.$$

If v is h -orthogonal to u , then

$$g^{(O)}(v, v) = h(v, v) > 0.$$

So one direction becomes timelike while the directions perpendicular to it stay spacelike. This is the mathematical core of the claim that **time is an entropy-selected direction**. Fisher–Rao by itself is positive-definite; the Obidi transformation is the extra step that makes relativistic signature possible. ²⁷

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Once the Obidi metric exists, one can compare its curvature to the curvature of the original information metric. Letter III introduces the corresponding correction schematically as

$$\Delta R = R[g^{(O)}] - R[h].$$

The physical meaning is that once an entropy-gradient direction is chosen and promoted to the role of time, the curvature is no longer the curvature of “raw information space”; it is the curvature of a time-oriented spacetime geometry. In this sense, the Obidi curvature correction measures the price of turning distinguishability geometry into relativistic geometry. filecite turn0file0

Letter III also proposes an **Obidi Curvature Invariant** with a lower bound tied to

$$\ln 2.$$

This is a very important place to be precise. The number $\ln 2$ is physically famous because one bit of binary information carries an entropy scale of $\ln 2$, and Landauer’s principle ties erasure of one bit to an energy cost $k_B T \ln 2$. But the specific object called the Obidi Curvature Invariant is **not** a standard invariant from mainstream differential geometry. It is a **ToE proposal** that uses the one-bit entropy quantum as a lower distinguishability floor. That makes conceptual sense inside entropic physics, but it should not be confused with a standard theorem about Riemannian curvature. ²⁸ filecite turn0file0

Now for the right-hand side of Einstein’s equation. Letter III gives a prototype **entropic source tensor** using the entropy field S . A canonical scalar-field-like example is

$$T_{\mu\nu}^{(S)} = \chi \nabla_\mu S \nabla_\nu S - g_{\mu\nu} \left[\frac{\chi}{2} \nabla^\alpha S \nabla_\alpha S - V(S) \right].$$

This formula is tremendously helpful for intuition. If the entropy field depends mainly on time, then in a comoving frame it produces an effective energy density and pressure

$$\rho_S = \frac{\chi}{2} \dot{S}^2 + V(S), \quad p_S = \frac{\chi}{2} \dot{S}^2 - V(S).$$

So even one entropic scalar can already behave like different kinds of matter depending on whether kinetic or potential terms dominate. That shows concretely how mass-energy and pressure can arise as coarse-grained information content. ToE then expands this idea by adding geometric and current sectors, so that the full dressed source is not just $T_{\mu\nu}^{(S)}$ but a sum of matter, entropy, and additional geometric-information contributions. [filecite turn0file0](#)

The dressed field equation in Letter III has the schematic form

$$G_{\mu\nu} + \Lambda_{\text{ent}} g_{\mu\nu} = 8\pi G_{\text{eff}} \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(S)} + T_{\mu\nu}^{(G)} \right) + \text{higher-information corrections.}$$

This is the mathematical statement of the ToE program. The left-hand side is generated from information geometry by emergence and Obidi-signature conversion; the right-hand side is generated from localized entropic sources and their coarse-grained moments. In a low-coupling, low-curvature, near-equilibrium limit, Letter III claims that one recovers ordinary Einstein gravity, and it also presents Bianconi's "gravity from entropy" as an intermediate or compatible entropic limit. ²⁰ [filecite turn0file0](#)

Letter III's irreversible physics appears in its **Vuli-Ndlela path integral** and in the split of the entropic field into reversible and irreversible sectors. Schematically, it writes

$$Z_{\text{ToE}} = \int \mathcal{D}S \exp \left[\frac{i}{\hbar_{\text{eff}}} (S_{\text{class}}[S] + S_G[S] + S_{\text{irr}}[S]) \right],$$

and also studies equations such as

$$\nabla_\mu \nabla^\mu S - \frac{\partial V}{\partial S} + \frac{i}{\hbar} \frac{\delta S_{\text{irr}}}{\delta S} = J(x), \quad \nabla_\mu J_{\text{ent}}^\mu \geq 0.$$

You do not need to master path integrals to understand the idea. The idea is that histories are still summed over, but irreversible entropy production changes the weighting of those histories. That is how ToE tries to encode the arrow of time mathematically rather than only philosophically. [filecite turn0file0](#)

A final point worth stating in the simplest possible language is this. In ToE, the Einstein right-hand side is not thrown away. It is **reinterpreted**. Pressure, stress, radiation, and mass are still there, but they are read as stable, localized, transportable information patterns on an entropic manifold. The goal is not to erase matter, but to derive matter's gravitational role from deeper informational tensors and currents. Whether that program can be completed in full detail is still open, but the mathematical path laid out in Letter III is coherent: moments $\rightarrow$ tensors, divergences $\rightarrow$ metrics, entropy gradients $\rightarrow$ time direction, coarse-graining $\rightarrow$ material sources, Bianchi compatibility $\rightarrow$ conserved gravitational field equations. ²⁹ [filecite turn0file0](#)

References for further reading

The uploaded **Letter III and appendix** are the primary source for ToE-specific objects such as the Obidi transformation, the Obidi metric, the Obidi Curvature Invariant, the Hybrid Metric-Affine Space, the reversible/irreversible entropic split, and the dressed Einstein-limit program. [filecite turn0file0](#)

For classical information geometry, the most useful starting points from this research pass are N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer, *Information geometry and sufficient statistics*; James G. Dowty, *Chentsov's theorem for exponential families*; and Frank Nielsen, *Cramer-Rao Lower Bound and Information Geometry*. These explain why Fisher–Rao and the Amari–Čencov tensor are mathematically privileged. ⁹

For α -connections and generalized Fisher structures, see Martin Bauer, Alice Le Brigant, Yuxiu Lu, and Cy Maor, *The L^p -Fisher–Rao metric and Amari–Čencov α -connections*, and Shimpei Kobayashi and Yu Ohno, *A characterization of the alpha-connections on the statistical manifold of multivariate normal distributions*. ³⁰

For classical and quantum Fisher geometry and its relation to Fubini–Study geometry, see Paolo Facchi and collaborators, *Classical and Quantum Fisher Information in the Geometrical Formulation of Quantum Mechanics*, and Ran Cheng, *Quantum Geometric Tensor (Fubini–Study Metric) in Simple Quantum System: A pedagogical Introduction*. ¹⁵

For mixed-state geometry and the reduction of Bures-type structures to Fubini–Study in appropriate limits, see Xu-Yang Hou and collaborators, *Local geometry and quantum geometric tensor of mixed states*. ³¹

For quantum relative entropy, see Takahiro Sagawa, *Second Law-Like Inequalities with Quantum Relative Entropy: An Introduction*, and Henrik Wilming, Rodrigo Gallego, and Jens Eisert, *Axiomatic characterization of the quantum relative entropy and free energy*. ¹⁸

For entropic or thermodynamic routes to gravity, see Ted Jacobson, *Thermodynamics of Spacetime: The Einstein Equation of State*; Erik Verlinde, *On the Origin of Gravity and the Laws of Newton*; and Ginestra Bianconi, *Gravity from entropy* together with her follow-up thermodynamic work. ³²

For the link between stress–energy, entropy current, and covariant statistical mechanics, see F. Becattini, *Covariant statistical mechanics and the stress-energy tensor*. For radiation moments in relativistic settings, see Masaru Shibata and collaborators, *Truncated Moment Formalism for Radiation Hydrodynamics in Numerical Relativity*. ⁸

For the physical significance of the one-bit entropy unit $\ln 2$, see the modern experimental literature on Landauer's principle, including Yonggun Jun, Momčilo Gavrilov, and John Bechhoefer, *High-precision test of Landauer's principle in a feedback trap*, and J. Hong and collaborators, *Experimental verification of Landauer's principle in erasure of nanomagnetic memory bits*. ²⁸

Open questions and limitations

Some of the most important objects in your request are **standard** and already well established: Fisher–Rao, Fubini–Study, KL divergence, Umegaki quantum relative entropy, α -connections, tangent and cotangent bundles, and the Bianchi identity. Others are **specific proposals inside ToE**: the Obidi transformation, the Obidi metric, the OCI, the α - q constitutive law in the exact form used by Letter III, and the staged derivation of the Einstein right-hand side from entropic tensors and currents. Those ToE objects should be read as original theoretical constructions presented in Letter III, not as settled consensus mathematics. ¹²

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The main mathematical challenge still open for ToE is not the intuition but the full proof chain: making the emergence map, the Lorentzian-signature conversion, the entropic source tensor, and the conservation laws all work globally and rigorously in one self-contained formalism. Letter III gives a program and many ingredients for that program. Whether the complete derivation can be closed without extra assumptions remains a live research question. ³³  

¹ ² ⁹ ¹⁰ ¹³ ²⁷ <https://arxiv.org/abs/1207.6736>
<https://arxiv.org/abs/1207.6736>

³ ³² <https://arxiv.org/abs/gr-qc/9504004>
<https://arxiv.org/abs/gr-qc/9504004>

⁴ ⁵ <https://arxiv.org/abs/1204.2191>
<https://arxiv.org/abs/1204.2191>

⁶ ⁷ <https://arxiv.org/abs/1604.07862>
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⁸ ²⁴ ²⁹ <https://arxiv.org/abs/1201.5278>
<https://arxiv.org/abs/1201.5278>

¹¹ <https://arxiv.org/abs/2303.07673>
<https://arxiv.org/abs/2303.07673>

¹² <https://arxiv.org/abs/1701.08895>
<https://arxiv.org/abs/1701.08895>

¹⁴ ³⁰ <https://arxiv.org/abs/2306.14533>
<https://arxiv.org/abs/2306.14533>

¹⁵ ²⁵ <https://arxiv.org/abs/1009.5219>
<https://arxiv.org/abs/1009.5219>

¹⁶ <https://arxiv.org/abs/1012.1337>
<https://arxiv.org/abs/1012.1337>

¹⁷ ³¹ <https://arxiv.org/abs/2305.07597>
<https://arxiv.org/abs/2305.07597>

¹⁸ ¹⁹ <https://arxiv.org/abs/1202.0983>
<https://arxiv.org/abs/1202.0983>

²⁰ ³³ <https://arxiv.org/abs/2408.14391>
<https://arxiv.org/abs/2408.14391>

²¹ ²³ <https://arxiv.org/abs/2201.00938>
<https://arxiv.org/abs/2201.00938>

²² <https://arxiv.org/abs/1512.06761>
<https://arxiv.org/abs/1512.06761>

²⁶ <https://arxiv.org/abs/2107.11925>
<https://arxiv.org/abs/2107.11925>

²⁸ <https://arxiv.org/abs/1408.5089>
<https://arxiv.org/abs/1408.5089>